

Test Cases

xUnit Test Suite — ToolForge Agent in C#

Team: ToolForge | Framework: xUnit (.NET 8) | Tests run offline (no API keys needed)

TEST
CASES

■ Test Suite Summary

File	Class	Tests	Coverage Area
AgentLoopTests.cs	AgentLoopTests	6	ReAct loop routing, termination, chaining, max-iteration guard
AgentToolTests.cs	TimeToolTests	4	TimeTool: valid/invalid/default timezone
AgentToolTests.cs	SqliteToolTests	6	SqliteTool: SELECT, empty result, DROP blocked, DELETE blocked
AgentToolTests.cs	WeatherToolTests	3	WeatherTool: empty city throws, live tests skipped
SqliteToolTests.cs	SqliteToolTests (ext)	14	Full SQL guard, injection, edge cases, seed data correctness
	TOTAL	33	All tools + agent loop

Note: All AgentLoop tests use **FakeHttpHandler** and **FakeToolRegistry** — zero real API calls are made. The test suite runs fully offline with dotnet test tests/.

■ AgentLoopTests.cs — ReAct Agent Loop

AL-00		
1	Loop_StopsImmediately_WhenFinishReasonIsStop	Happy Path PASS
Description:	When the API responds with finish_reason=stop on the first turn, the loop terminates immediately and returns the assistant's reply text.	
Input:	Hi there	
Expected:	Reply = "Hello from ToolForge!" ToolCallCount = 0	

AL-00 2	Loop_InvokesTool_ThenReturnsAnswer	Tool Call	PASS
Description:	When the API returns finish_reason=tool_calls, the loop dispatches to the correct tool, appends the result, and loops until finish_reason=stop.		
Input:	What time is it in India?		
Expected:	Reply contains "10:30" ToolCallCount = 1 get_time was called		
AL-00 3	Loop_CanChainMultipleTools	Multi-Tool	PASS
Description:	The agent correctly chains two sequential tool calls (get_weather then get_time) for a compound query, accumulating both results before generating the final answer.		
Input:	What is the weather and time in Chennai?		
Expected:	ToolCallCount = 2 get_weather called get_time called		
AL-00 4	Loop_RespectsMaxIterationLimit	Guard	PASS
Description:	If the API keeps returning tool_calls indefinitely, the loop self-terminates after 5 iterations and returns a 'Max iterations reached' message.		
Input:	Weather? (API returns infinite tool_calls)		
Expected:	Reply contains "Max iterations" No exception thrown		
AL-00 5	Loop_HandlesToolError_Gracefully	Error Handling	PASS
Description:	When a tool returns an error string (not an exception), the agent passes it back to the model which generates a user-friendly explanation — no crash.		
Input:	Weather in Xyz123?		
Expected:	ToolCallCount = 1 Reply does not contain 'Exception'		
AL-00 6	PromptBuilder_SystemPrompt_ContainsAllTools	Prompt	PASS
Description:	The system prompt returned by PromptBuilder.SystemPrompt() contains documentation for all three registered tools.		
Input:	(no input — static method)		
Expected:	Contains: get_weather, get_time, query_sqlite		

■ TimeToolTests — TimeTool.cs

ID	Test Method	Input	Expected Output	Result
TT-001	GetTime_ValidTimezone_ReturnsFormattedString	<code>get_time("Asia/Kolkata")</code>	Non-empty string containing 'IST'	PASS
TT-002	GetTime_UTC_ReturnsUtcTime	<code>get_time("UTC")</code>	String containing 'UTC'	PASS
TT-003	GetTime_InvalidTimezone_ReturnsErrorString	<code>get_time("Mars/Olympus")</code>	Contains 'Unknown timezone', not 'Exception'	PASS
TT-004	GetTime_DefaultsToUtc	<code>get_time()</code> (no args)	String containing 'UTC'	PASS

WeatherToolTests — WeatherTool.cs

Tests WT-002 and WT-003 require a live OPENWEATHER_API_KEY and are decorated with `[Fact(Skip = ...)]` so they do not run in CI without the key set.

ID	Test Method	Input	Expected	Result
WT-001	GetWeather_EmptyCity_ThrowsArgumentException	<code>""</code>	ArgumentException thrown	PASS
WT-002	GetWeather_ValidCity_ReturnsTemperature	<code>"Chennai"</code>	Contains 'Temp:' and 'Humidity:'	SKIP*
WT-003	GetWeather_InvalidCity_ReturnsUserFriendlyError	<code>"ZZZInvalidCity999"</code>	Contains 'not found', no exception	SKIP*

* Skipped — requires live API key. Set OPENWEATHER_API_KEY to enable.

SqliteToolTests — SqliteTool.cs (Extended Suite)

ID	Method	Input	Expected	Result	Type
ST-001	Query_SelectAll_ReturnsJsonArray	<code>SELECT * FROM employees LIMIT 1</code>	Starts with '[', contains 'name'	PASS	Happy
ST-002	Query_TopSalespeople_ReturnsCorrectOrder	<code>SELECT name, sales ... ORDER BY sales DESC</code>	Priya appears before Arjun in output	PASS	Happy
ST-003	Query_ProductsByCategory_FiltersCorrectly	<code>SELECT ... WHERE category='Electronics'</code>	Contains Mouse, not Standing Desk	PASS	Happy
ST-004	Query_AggregateCount_Works	<code>SELECT COUNT(*) as total FROM employees</code>	Contains 'total' and '7'	PASS	Happy

ID	Method	Input	Expected	Result	Type
ST-005	Query_JoinLike_CrossTableFilter_Works	SELECT ... WHERE dept='Sales'	Contains Priya & Arjun, not Vikram	PASS	Happy
ST-006	Query_EmptyResultSet_ReturnsNoResultsMessage	SELECT ... WHERE name='__nobody__'	"No results found."	PASS	Edge
ST-007	Query_DropStatement_IsBlocked	DROP TABLE employees	Contains 'Blocked'	PASS	Security
ST-008	Query_DeleteStatement_IsBlocked	DELETE FROM employees	Contains 'Only SELECT'	PASS	Security
ST-009	Query_DestructiveSQL_IsBlocked (Theory)	INSERT / UPDATE / ALTER / TRUNCATE variants	Blocked message + employee count still 7	PASS	Security
ST-010	Query_SelectWithEmbeddedDrop_IsBlocked	SELECT *; DROP TABLE employees; --	Contains 'Blocked' or 'Only SELECT'	PASS	Security
ST-011	Query_AttachKeyword_IsBlocked	ATTACH DATABASE '/etc/passwd'	Contains 'Only SELECT'	PASS	Security
ST-012	Query_EmptyString_ReturnsEmptyError	""	Contains 'empty'	PASS	Edge
ST-013	Query_SyntaxError_ReturnsUserFriendlyError	SELECT FROM WHERE	Contains 'failed', not 'Unhandled'	PASS	Edge
ST-014	Query_DoesNotThrow_ReturnsStringAlways	Various malformed SQL strings (3 variants)	No exception thrown for any input	PASS	Contract

■ How to Run the Test Suite

```
# From the project root cd tests dotnet test --verbosity normal # Expected output Test run for tests.dll (.NETCoreApp,Version=v8.0) Microsoft (R) Test Execution Command Line Tool Version 17.x ... Passed! - Failed: 0, Passed: 31, Skipped: 2, Total: 33
```

The 2 skipped tests are WeatherTool live API tests. All other 31 tests pass offline. Set **OPENWEATHER_API_KEY** in your environment to enable the skipped tests.